

COMS/SE 4170

Software Testing

Spring 2026

Combination Strategies
(Chapters 6.2-6.4)

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ISP Criteria – Base Choice

- Testers sometimes recognize that certain values are important

Base Choice Coverage (BCC) : A base choice block is chosen for each characteristic, and a base test is formed by using the base choice for each characteristic. Subsequent tests are chosen by holding all but one base choice constant and using each non-base choice in each other characteristic.

- Number of tests is one base test + one test for each other block $1 + \sum_{i=1}^Q (B_i - 1)$

For triang() : Base A1, B1, C1

A1, B1, C2	A1, B2, C1	A2, B1, C1
A1, B1, C3	A1, B3, C1	A3, B1, C1
A1, B1, C4	A1, B4, C1	A4, B1, C1

Introduction to Software Testing, Edition 2 (Ch 6)

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Base Choice

simplified PowerPoint

Parameters:

Open_gallery:

on.

off.

Turn_off_DM:

on.

off.

Print_Quality:

high.

med.

Pen:

on.

off.

Custom_Actions:

do_not_allow.

allow_with.

allow_without.

Base:

OG: -on

Toff: -on

Print: med

Pen: off

Custom: don't allow

N=1+(1+1+1+1+2)=7

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Parameters:

Open_gallery:

on.

off.

Turn_off_DM:

on.

off.

Print_Quality:

high.

med.

Pen:

on.

off.

do_not_allow.

allow_with.

allow_without.

OG	Toff	Print	Pen	custom
On	On	Med	Off	Don't allow
Off	On	Med	Off	Don't Allow
On	Off	Med	Off	Don't allow
On	On	High	Off	Don't' allow
On	On	Med	On	Don't allow
On	On	Med	Off	Allow-with
On	On	Med	Off	Allow-without

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Base Choice Notes

- The base **test must be feasible**
 - That is, all base choices must be compatible
- Base choices can be
 - Most likely from an end-use point of view
 - Simplest
 - Smallest
 - First in some ordering
- Happy path tests often make good base choices
- The base choice is a crucial design decision
 - Test designers should document why the choices were made

ISP Criteria – Multiple Base Choice

Multiple Base Choice Coverage (MBCC) : At least one, and possibly more, base choice blocks are chosen for each characteristic, and base tests are formed by using each base choice for each characteristic at least once. Subsequent tests are chosen by holding all but one base choice constant for each base test and using each non-base choice in each other characteristic.

- If M base tests and m_i base choices for each characteristic:

$$M + \sum_{i=1}^Q (M * (B_i - m_i))$$

For *triang()* : Bases

A1, B1, C1	A1, B1, C3	A1, B3, C1	A3, B1, C1
	A1, B1, C4	A1, B4, C1	A4, B1, C1
A2, B2, C2	A2, B2, C3	A2, B3, C2	A3, B2, C2
	A2, B2, C4	A2, B4, C2	A4, B2, C2

ISP Coverage Criteria Subsumption

